EE4-66
EE9-SO28

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

TOPICS IN LARGE DIMENSIONAL DATA PROCESSING

Friday, 15 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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EE4-66 Topics in Large Dimensional Data Processing

Instructions for Candidates

Answer all three questions. Total marks: 75.

1. (Linear Inverse Problem)

   (a) Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a flat matrix, i.e., $m < n$.

      i Consider the full singular value decomposition (SVD) of $\boldsymbol{A}$ in the form of $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$. Identify the dimensions and properties of the matrices $\boldsymbol{U}$, $\boldsymbol{\Sigma}$, and $\boldsymbol{V}$, respectively. [3]

      ii Let r be the (row) rank of $\boldsymbol{A}$. Suppose that $r \leq m < n$. The compact SVD of $\boldsymbol{A}$ is given by $\boldsymbol{A} = \boldsymbol{U}_r\boldsymbol{\Sigma}_r\boldsymbol{V}_r^T$.

         A. Identify the dimensions of the matrices $\boldsymbol{U}_r$, $\boldsymbol{\Sigma}_r$, and $\boldsymbol{V}_r$, respectively. [2]

         B. Describe the relationship between $\boldsymbol{U}_r, \boldsymbol{\Sigma}_r, \boldsymbol{V}_r$ and $\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$. [2]

      iii Decompose the matrices $\boldsymbol{A}\boldsymbol{A}^T$ and $\boldsymbol{A}^T\boldsymbol{A}$ and write your results in terms of $\boldsymbol{U}_r$, $\boldsymbol{\Sigma}_r$, and $\boldsymbol{V}_r$. [2]

      iv Define the pseudo-inverse of $\boldsymbol{A}$, denoted by $\boldsymbol{A}^\dagger$, using $\boldsymbol{U}_r$, $\boldsymbol{\Sigma}_r$, and $\boldsymbol{V}_r$. [2]

      v For a given arbitrary $\boldsymbol{y} \neq \boldsymbol{0}$ and $\boldsymbol{y} \in \text{range}(\boldsymbol{U}_r)$, it is clear that the set defined as

$$\mathcal{X} := \{\boldsymbol{x} : \ \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}\}$$

      is not empty.

         A. Let $\boldsymbol{x}_0 = \boldsymbol{A}^\dagger\boldsymbol{y}$ where $\boldsymbol{A}^\dagger$ is the psuedo-inverse of $\boldsymbol{A}$ defined in 1.(a)-iv. Show that $\boldsymbol{x}_0 \in \mathcal{X}$. [2]

         B. Define

$$\mathcal{X}_\perp := \{\boldsymbol{x}_\perp : \ \boldsymbol{A}\boldsymbol{x}_\perp = \boldsymbol{0}\}.$$

         It is clear that any $\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{x}_\perp$, where $\boldsymbol{x}_\perp \in \mathcal{X}_\perp$, is a solution of $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$.
         Find the dimension of the subspace $\mathcal{X}_\perp$ and an orthonormal basis of it via the SVD of $\boldsymbol{A}$. [2]

         C. It is a well-known fact that for all $\boldsymbol{x} \in \mathcal{X}$, it holds $\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{x}_\perp$ for some $\boldsymbol{x}_\perp \in \mathcal{X}_\perp$.
         Prove that

$$\|\boldsymbol{x}_0\|_2^2 = \min_{\boldsymbol{x} \in \mathcal{X}} \|\boldsymbol{x}\|_2^2.$$

         [2]

(b) Let

$$\boldsymbol{A} = \begin{bmatrix} 2 & 2 & 2 & 2 \\ 1 & 1 & -1 & -1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

i Compute $\boldsymbol{A}\boldsymbol{A}^T$. [2]

ii Let $\boldsymbol{v} = \frac{1}{2}[1, -1, 1, -1]^T$. Compute $\|\boldsymbol{v}\|_2$ and $\boldsymbol{A}\boldsymbol{v}$. [2]

iii The *full* SVD of $\boldsymbol{A}$ can be written as $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$. Find $\boldsymbol{U}$, $\boldsymbol{\Sigma}$, and $\boldsymbol{V}$ numerically. [2]

iv Find the pseudo-inverse of $\boldsymbol{A}$ numerically, denoted by $\boldsymbol{A}^\dagger$. [3]

v Let $\boldsymbol{y} = [4, 2, 1]^T$. Find $\boldsymbol{x} = \boldsymbol{A}^\dagger\boldsymbol{y}$ numerically. [2]

vi Let $\boldsymbol{x}'$ satisfy that $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}'$ and $\|\boldsymbol{x}'\|_0 = 1$. Find $\boldsymbol{x}'$ numerically. [2]

(Total marks: 30)

2. (Greedy Algorithms for Sparse Inverse Problems) Consider greedy algorithms to solve the following sparse inverse problem:

$$\min_{\boldsymbol{x}} \ \|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2^2 \ \text{s.t.} \ \|\boldsymbol{x}\|_0 \leq S,$$

where $\boldsymbol{y} \in \mathbb{R}^m$, $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, $\boldsymbol{x} \in \mathbb{R}^n$, and $0 < S \ll n$ is a given positive integer. We assume that all the columns of $\boldsymbol{A}$ are normalised, i.e., $\|\boldsymbol{a}_j\|_2 = 1$ where $\boldsymbol{a}_j$ is the j-th column of $\boldsymbol{A}$.

(a) The Orthogonal Matching Pursuit (OMP) algorithm is stated in Algorithm 2.1.

Algorithm 2.1 OMP Algorithm

1: **Input**: S, $\boldsymbol{A}$, and $\boldsymbol{y}$.
2: **Initialisation**: $\boldsymbol{x} = \boldsymbol{0}$, $\mathcal{T}^0 = \phi$, and $\boldsymbol{y}_r = \boldsymbol{y}$.
3: **for** $l = 1, 2, \cdots, S$ **do**
4: $\quad i_l = ??$ ▷ Find a new index
5: $\quad \mathcal{T}^l = \mathcal{T}^{l-1} \bigcup \{i_l\}$ ▷ Expand the support set $\mathcal{T}$
6: $\quad \boldsymbol{x}_{\mathcal{T}^l} = ??$ ▷ Update $\boldsymbol{x}$
7: $\quad \boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$ ▷ Update $\boldsymbol{y}_r$

i Lines 4 and 6 in OMP are incomplete. Complete them. [2]

ii Prove that generally speaking $i_l \notin \mathcal{T}^{l-1}$. [4]

iii The performance guarantee of OMP states that if the mutual coherence of the matrix $\boldsymbol{A}$ satisfies

$$\mu(\boldsymbol{A}) \leq \frac{1}{2S},$$

then OMP guarantees to recover an arbitrary S-sparse signal $\boldsymbol{x}_0$ exactly.

A. State the definition of mutual coherence $\mu(\boldsymbol{A})$. [1]

B. Let $\boldsymbol{x}_0$ be the ground-truth signal that generates $\boldsymbol{y}$, i.e., $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}_0$. Let $\mathcal{T}_0 = \text{supp}(\boldsymbol{x}_0)$ and $|\mathcal{T}_0| \leq S$. Show that for all $i \notin \mathcal{T}_0$, $|\langle \boldsymbol{a}_i, \boldsymbol{y}\rangle| \leq \mu\sqrt{S}\,\|\boldsymbol{x}_0\|_2$. [3]
Hint: Cauchy-Schwartz inequality $|\langle \boldsymbol{u}, \boldsymbol{v}\rangle| \leq \|\boldsymbol{u}\|_2 \|\boldsymbol{v}\|_2$ is needed.

(b) The Subspace Pursuit (SP) algorithm is stated in Algorithm 2.2.

i Complete Lines 4, 5, 6, and 7 of the SP algorithm. [2]

ii In the SP algorithm, H_S denotes the hard thresholding function with sparsity S. Find $H_3([-9, 8, -7, 6, -5]^T)$. [2]

Algorithm 2.2 SP Algorithm

Input: S, $\boldsymbol{A}$, and $\boldsymbol{y}$.
Initialisation: $\mathcal{T}^0 = \text{supp}(H_S(\boldsymbol{A}^T\boldsymbol{y}))$, and $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}_{\mathcal{T}^0}\boldsymbol{A}_{\mathcal{T}^0}^{\dagger}\boldsymbol{y}$.
while $l = 1, 2, \cdots$ until stop criteria hold **do**
    $\tilde{\mathcal{T}}^l = \mathcal{T}^{l-1} \bigcup ??$ ▷ Expand support
    Set $\boldsymbol{b}$ by ?? ▷ Estimate a $2S$-sparse signal
    Set $\mathcal{T}^l =??$ ▷ Shrink support
    Set $\boldsymbol{x}$ by ?? ▷ Estimate an S-sparse signal
    $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$ ▷ Update $\boldsymbol{y}_r$

iii The performance guarantees of SP is obtained through the so called restricted isometry property (RIP) and restricted isometry constant (RIC). State the definitions of RIP and RIC. [4]

iv Consider two submatrices of $\boldsymbol{A}$, given by $\boldsymbol{A}_{\mathcal{I}}$ and $\boldsymbol{A}_{\mathcal{J}}$ where $\mathcal{I}, \mathcal{J} \subset \{1, 2, \cdots, n\}$, $\mathcal{I} \bigcap \mathcal{J} = \phi$, and $|\mathcal{I}| = |\mathcal{J}| = S$. RIP suggests near-orthogonality of $\boldsymbol{A}_{\mathcal{I}}$ and $\boldsymbol{A}_{\mathcal{J}}$. Find the mathematical expression to state the near-orthogonality of $\boldsymbol{A}_{\mathcal{I}}$ and $\boldsymbol{A}_{\mathcal{J}}$ using singular values and RIC. Proof is not needed here. [2]

(Total marks: 20)

3. (Convex Optimisation, and Sparse Inverse Problems)

(a) Consider the following constrained optimisation problem

$$\min_{\boldsymbol{x}} \frac{1}{2} \|\boldsymbol{x}\|_2^2 \text{ s.t. } \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}, \tag{3.1}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ is the unknown variable vector to be optimised, $\boldsymbol{y} \in \mathbb{R}^m$ and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ are given.

i Let $\boldsymbol{v}$ be the corresponding Lagrange multiplier vector. Specify the dimension of $\boldsymbol{v}$ and find the Lagrangian of (3.1), denoted by $L(\boldsymbol{x}, \boldsymbol{v})$. [2]

ii Find the corresponding dual function $g(\boldsymbol{v})$. Solve the optimisation problem involved and obtain the closed form of $g(\boldsymbol{v})$. [2]

iii Find the form of the corresponding dual problem. Solve it by finding the optimal $\boldsymbol{v}^\star$. [2]

iv Find the closed form of the optimal solution $\boldsymbol{x}^\star$ of (3.1). [2]

(b) Consider the Lasso problem

$$\min_{\boldsymbol{x}} \frac{1}{2} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2 + \lambda \|\boldsymbol{x}\|_1, \tag{3.2}$$

where $\lambda > 0$ is a regularization constant.

i Consider an easier problem

$$\min_x f(x) := \min_x \frac{1}{2}(y - x)^2 + \lambda |x|,$$

where $x, y \in \mathbb{R}$. Find the derivative (or sub-differential) $\partial_x f(x)$. Then find the optimal solution $x^\star = \eta_\lambda(y)$ of the above scalar Lasso problem. [4]

In the literature, $\eta_\lambda(y)$ is referred to as the soft thresholding function.

ii Define

$$g(\boldsymbol{x}) := \frac{1}{2} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2.$$

For a fixed point $\boldsymbol{x}^k$, $g(\boldsymbol{x})$ can be approximated by

$$g(\boldsymbol{x}) \approx \frac{1}{2} \left\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^k\right\|_2^2 + \left\langle \nabla g\left(\boldsymbol{x}\right)|_{\boldsymbol{x}^k}, \boldsymbol{x} - \boldsymbol{x}^k \right\rangle + \frac{1}{2t_k} \left\|\boldsymbol{x} - \boldsymbol{x}^k\right\|_2^2, \tag{3.3}$$

for some properly chosen $t_k > 0$ when $\boldsymbol{x}$ is sufficiently close to $\boldsymbol{x}^k$.

Use the approximation (3.3) to derive the Iterative Shrinkage Thresholding (IST) algorithm to solve the Lasso problem (3.2). Use the soft thresholding function in your final solution. [3]

(c) Consider the constrained optimisation problem

$$\min_{\boldsymbol{x}} \ \|\boldsymbol{x}\|_1 \ \text{s.t.} \ \boldsymbol{y} = \boldsymbol{D}\boldsymbol{x}, \tag{3.4}$$

where $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{y} \in \mathbb{R}^m$, and $\boldsymbol{D} \in \mathbb{R}^{m \times n}$.

We will derive an ADMM algorithm to solve (3.4).

Generally speaking, ADMM is designed to solve convex optimisation problem in the following standard form

$$\begin{aligned} &\min_{\boldsymbol{x},\boldsymbol{z}} \ f(\boldsymbol{x}) + g(\boldsymbol{z}) \\ &\text{s.t.} \ \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} = \boldsymbol{c}. \end{aligned} \tag{3.5}$$

In high level, ADMM solves the above optimisation problem by solving much easier sub-problems iteratively.

i Consider the optimisation problem

$$\begin{aligned} &\min_{\boldsymbol{x},\boldsymbol{z}} \ \|\boldsymbol{z}\|_1 \\ &\text{s.t.} \ \boldsymbol{D}\boldsymbol{x} = \boldsymbol{y}, \ \text{and} \ \boldsymbol{x} = \boldsymbol{z}. \end{aligned} \tag{3.6}$$

Clearly (3.6) and (3.4) are equivalent. The advantage of (3.6) is that sub-problems in ADMM will be easy to solve. Specify f, g, $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{c}$ when mapping (3.6) directly to the ADMM standard form (3.5). [3]

ii Find the augmented Lagrangian $L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{v})$. Use $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{c}$ to simplify the notations. Find the dimension of the Lagrange multiplier vector $\boldsymbol{v}$. [3]

iii There are three steps in each iteration of ADMM:

$$\boldsymbol{x}^{k+1} = \arg\min_{\boldsymbol{x}} \ L_\rho(\boldsymbol{x}, \boldsymbol{z}^k, \boldsymbol{v}^k), \tag{3.7}$$

$$\boldsymbol{z}^{k+1} = \arg\min_{\boldsymbol{z}} \ L_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{z}, \boldsymbol{v}^k), \tag{3.8}$$

$$\boldsymbol{v}^{k+1} = \boldsymbol{v}^k + \rho(\boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\boldsymbol{z}^{k+1} - \boldsymbol{c}).$$

Find the closed-form solutions of (3.7) and (3.8) respectively. [4]

(Total marks: 25)